# Energy Loss as a Probe of Quark-Gluon Plasma Formation Across Collision System Size

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Based on CF and W. A. Horowitz, arXiv:2505.14568 [hep-ph] (2025); and
CF and W. A. Horowitz, Phys. Lett. B **864**, 139437 (2025).

NITheCS
National Institute for
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SPES BONA

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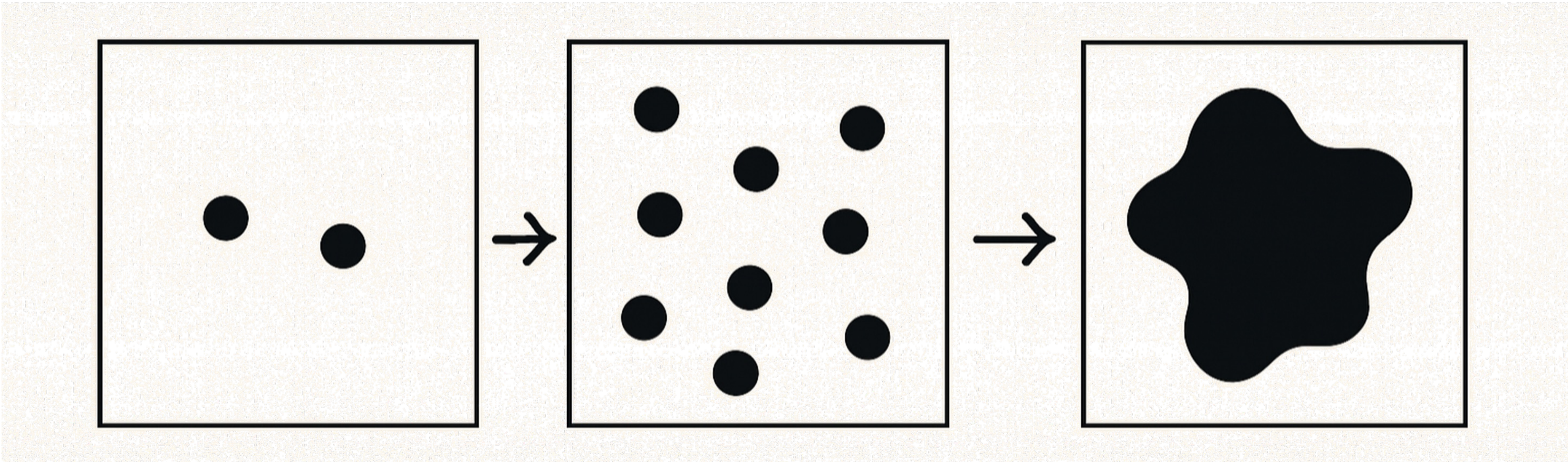
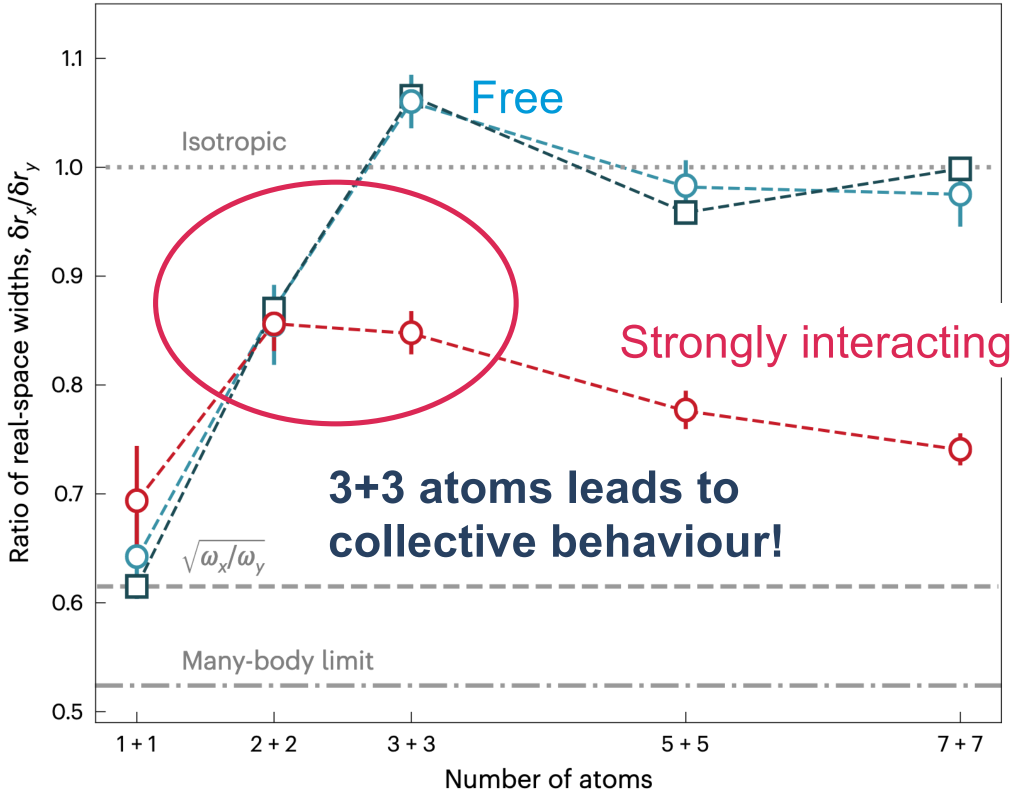
NRF National Research Foundation

# How many particles for collectivity?



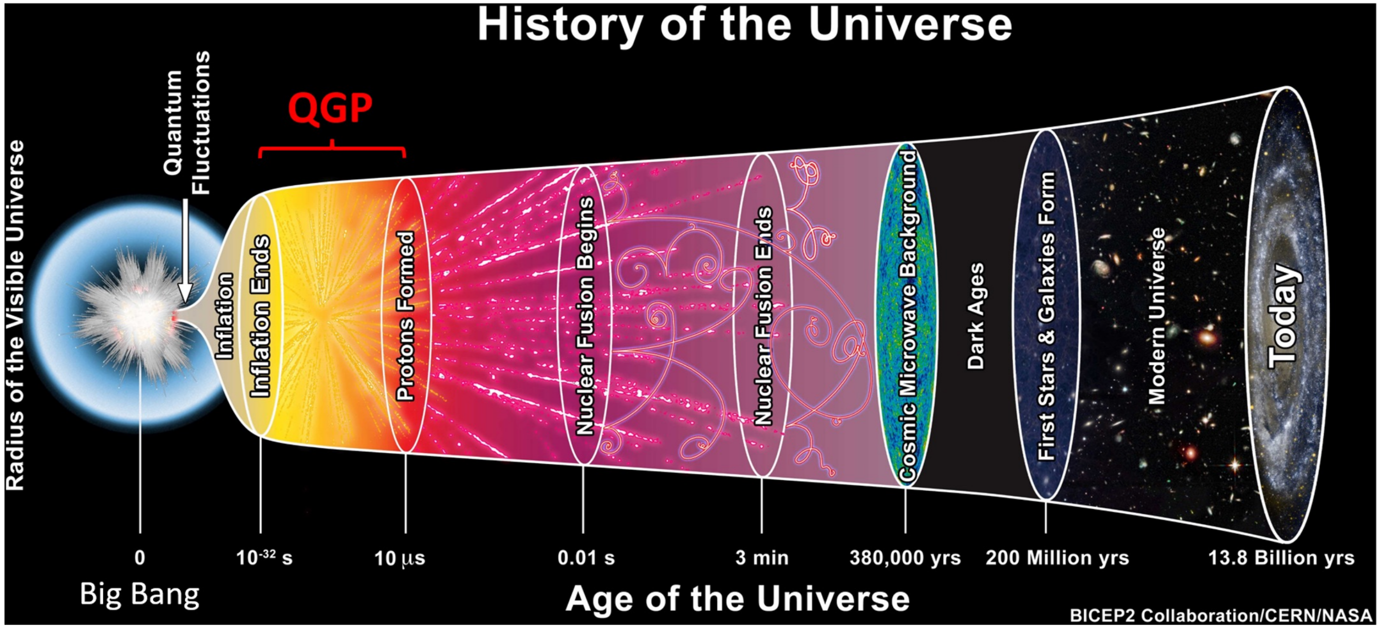
# Surprising Lessons from Cold Atoms

***Cold atom experiments*** **indicate that the answer is 6!**




S. Brandstetter, et al., Nat. Phys. **21**, 1 (2025).

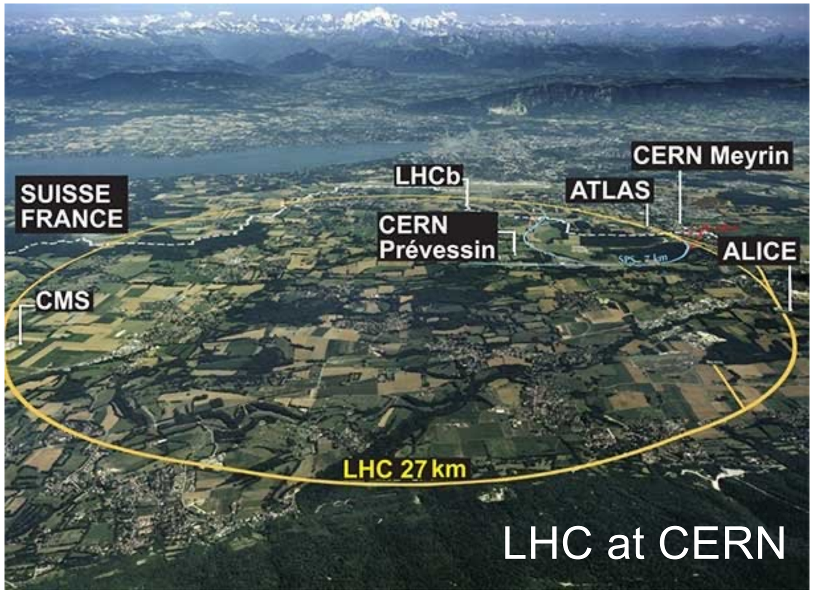
# Turning up the Temperature

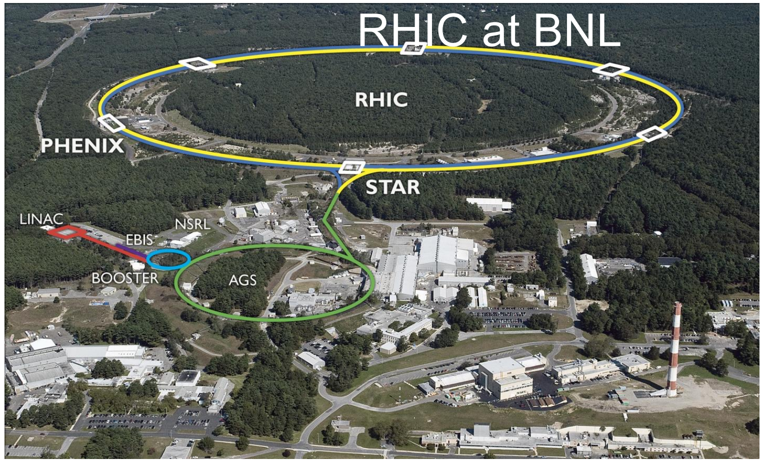



The early universe was made up of a dense, hot soup of quarks and gluons: the **quark gluon plasma**

# Heavy-ion collisions

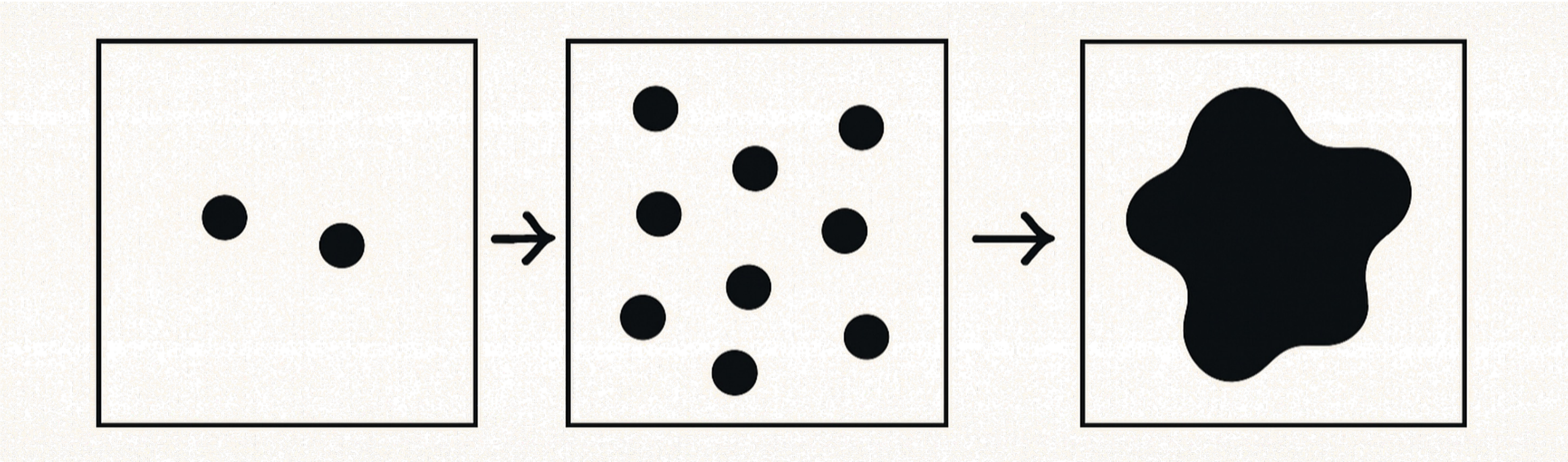
>99.9999% of the speed of light



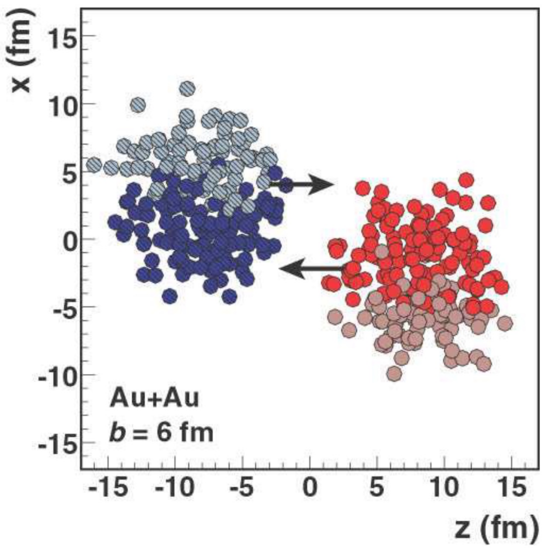



# How many particles for collectivity?

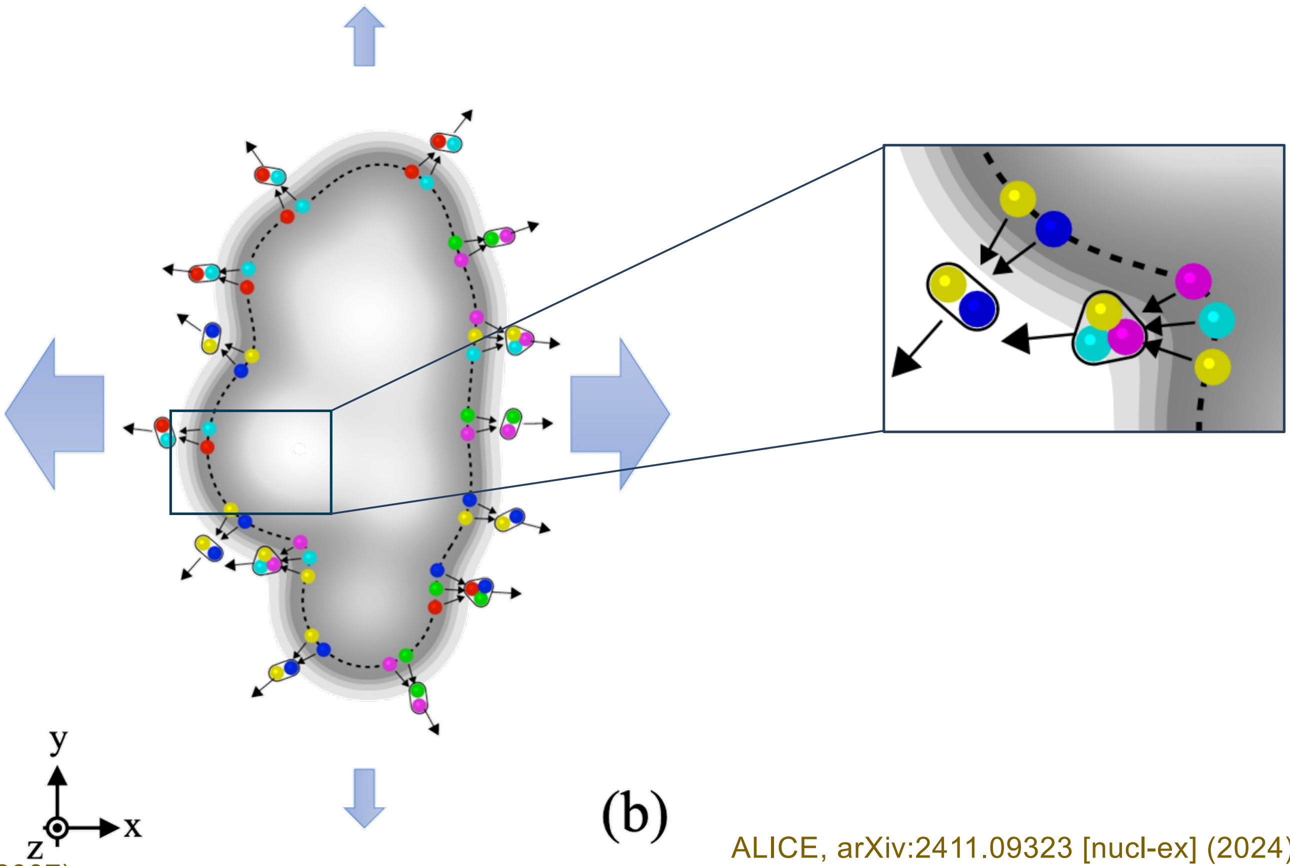
From statistical mechanics, you might say ~ 1 mole ~ $10^{23}$



# Lessons from Heavy-Ion Physics




(a)



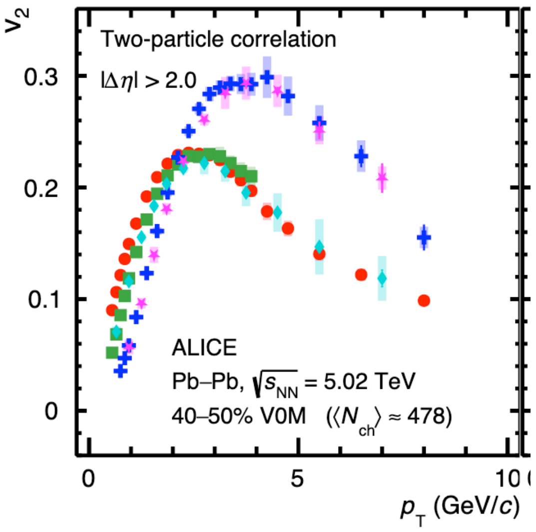

(b)

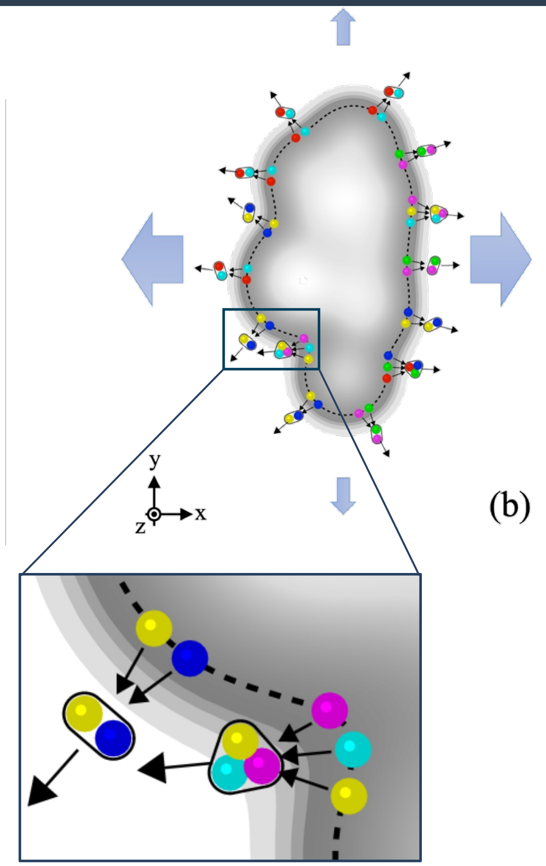
M. L. Miller, K. Reygers, S. J. Sanders, and P. Steinberg, Ann. Rev. Nucl. Part. Sci. **57**, 205 (2007).

ALICE, arXiv:2411.09323 [nucl-ex] (2024).

# How many particles for (hot) collectivity?

Pb + Pb

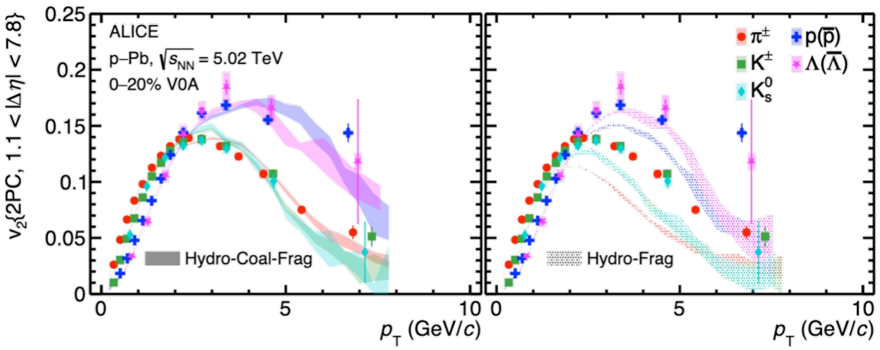



- Baryon-meson **grouping** and **splitting**
- ⇒ Flowing medium of quarks and gluons!

ALICE, arXiv:2411.09323 [nucl-ex] (2024).

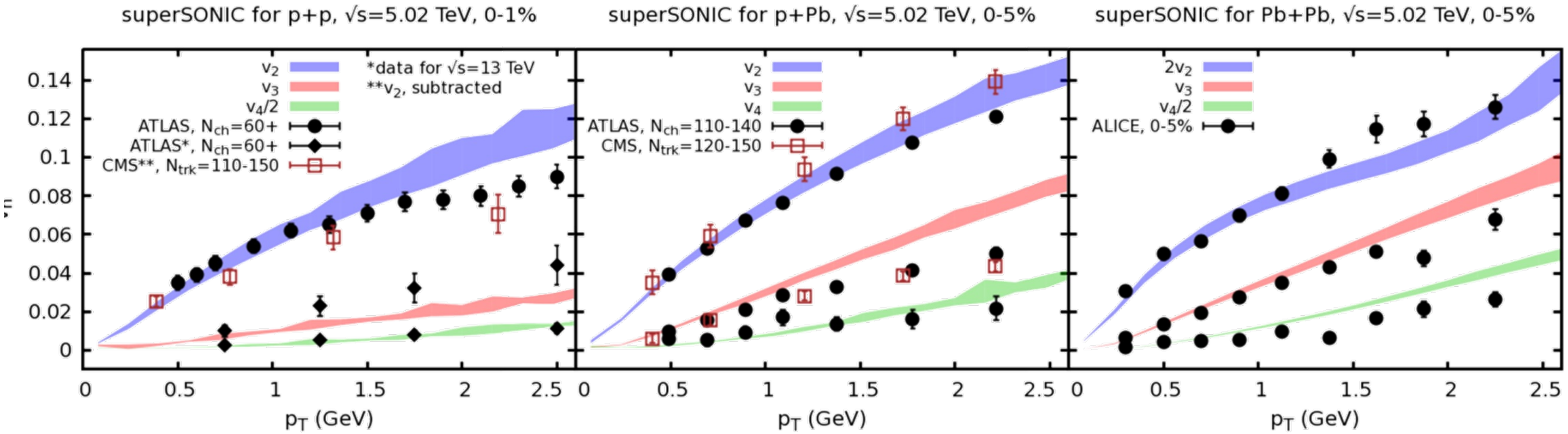
# Theory describes low-$p_T$ in small sys.




The low-$p_T$ physics is well-described by hydrodynamic models that include coalescence

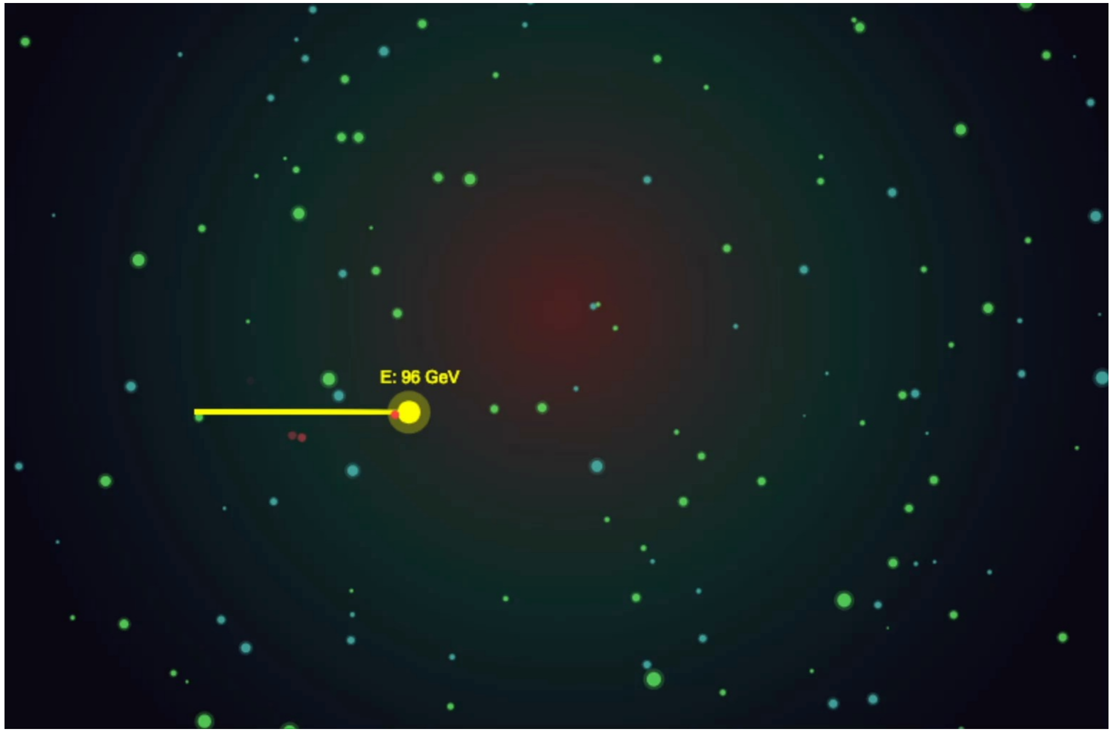
ALICE, arXiv:2411.09323 [nucl-ex] (2024).

Weller et al., PLB 774 (2017) 351-356

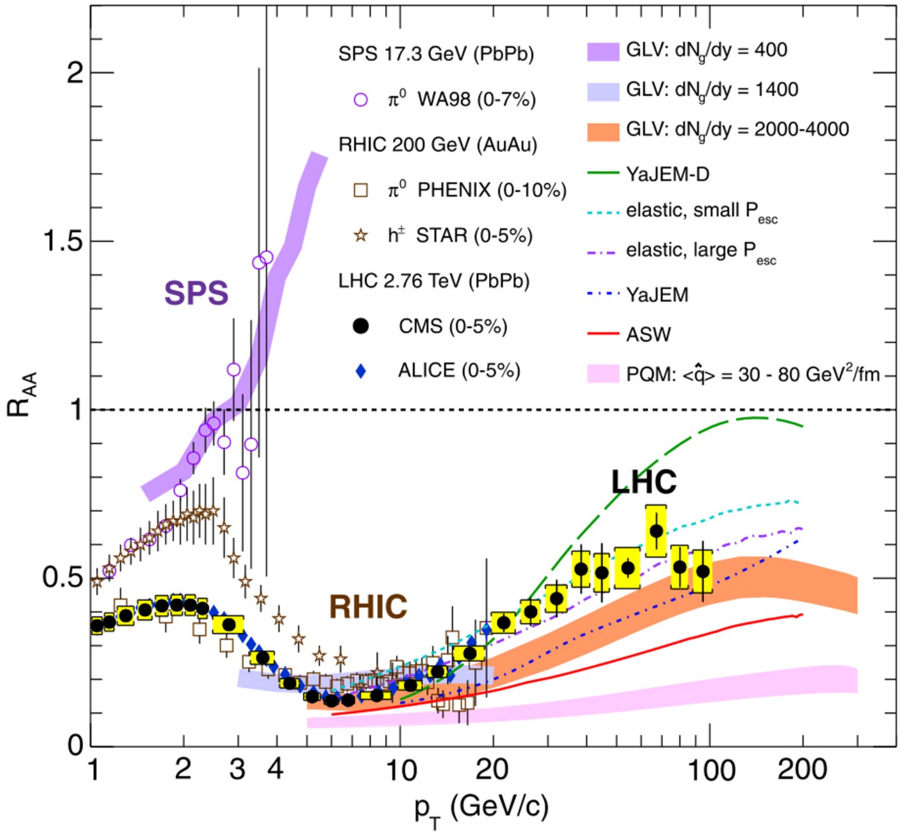


# Energy Loss

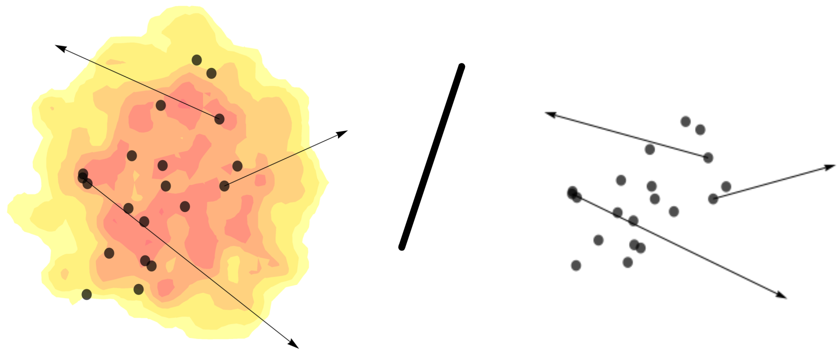
One can also probe the QGP by measuring energy loss of high momentum particles



# Measuring Energy Loss




$$R_{AA}(p_T) = \frac{1}{\langle N_{\text{coll}} \rangle} \frac{dN_{AA}/dp_T}{dN_{pp}/dp_T}$$

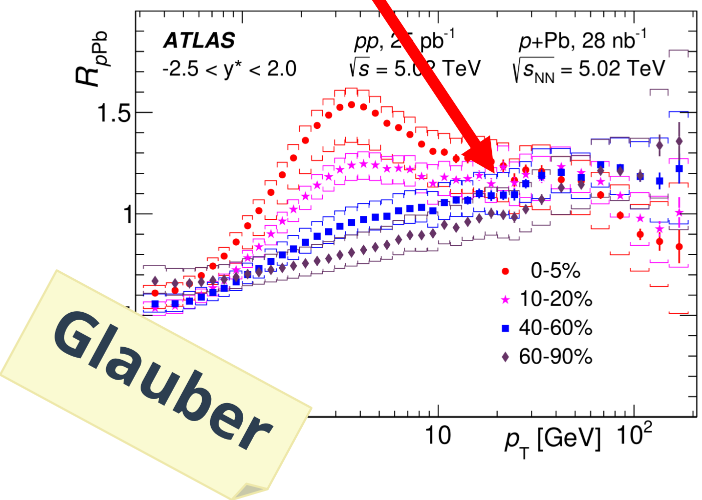
$=$ 

Quantifies how much **energy is lost** by high momentum particles

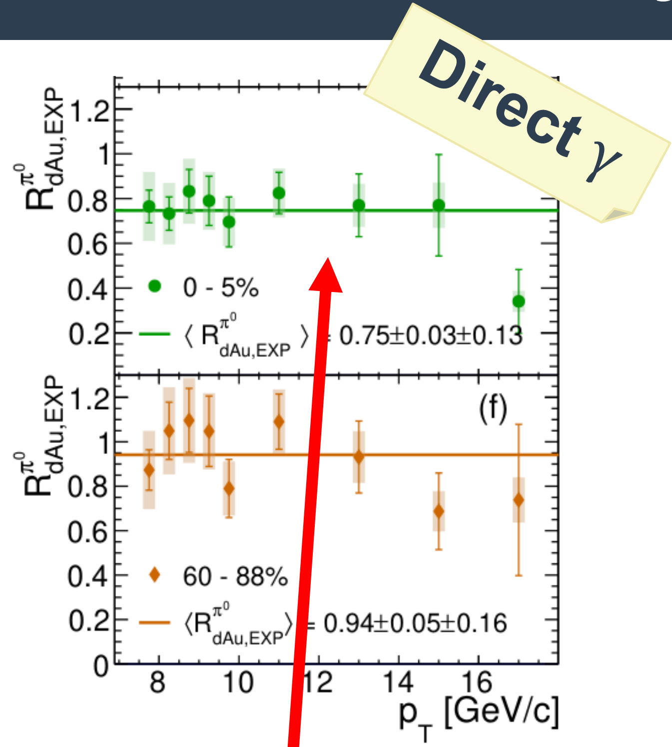
# Nuclear Modification in Small Systems

**Small system suppression pattern not clear**

No suppression!




*ATLAS, JHEP 07 (2023) 074*




Suppression ~0.75!

*PHENIX, Phys. Rev. Lett.* ***134****, 022302 (2025).*

- Apparent tension between RHIC and LHC suppression results?

→ **Theoretical input needed**

| Observable or effect | Pb– $A+A$ | p–P $p+A$ | pp $p+p$ | Refs. |
|---|---|---|---|---|
| Near-side ridge yields | ✅ | ✅ | ✅ | 1–8,8,9 |
| Azimuthal anisotropy | ✅ $v_1$–$v_9$ | ✅ $v_1$–$v_5$ | ✅ $v_2$–$v_4$ | 2–8,10–25 |
| **Flow** …ce | ✅ | ✅ | ✅ | 16,26–34 |
| …ss dependence | ✅ $v_2$–$v_5$ | ✅ $v_2$, $v_3$ | ✅ $v_2$ | 12,15,17,21,35–42 |
| Higher-order cumulants (mainly $v_2\{n\}$, $n \geq 4$) | ✅ "$4 \approx 6 \approx 8 \approx$ LYZ" +higher harmonics | ✅ "$4 \approx 6 \approx 8 \approx$ LYZ" +higher harmonics | ✅ "$4 \approx 6$" | 17,18,23,32,41,43–55 |
| Symmetric cumulants (SC) | ✅ up to $(5,3)$ | ✅ only $(4,2)$, $(3,2)$ | ✅ only $(4,2)$, $(3,2)$ | 20,23,56–62 |
| Non-linear flow modes | ✅ up to $v_7$ | ? | ? | 24,63,64 |
| **Correlations** | ✅ $n = 2$–$4$, $\{2\}, \{4\}$ | ✅ $n = 2, 3$, $\{2\}$ | ? | 7,19,65–69 |
| | ✅ $v_2$–$v_4$ | ? | ? | 70–72 |
| | ✅ up to $v_4$ | ✅ $v_2$ | ? | 73,74 |
| Directed flow (from spectators) | ✅ | ? | ? | 75 |
| Charge-dependent correlations | ✅ | ✅ | ✅ | 76–82 |
| **Freezeout properties** | ✅ | ✅ | ✅ | 83–94 |
| …on") | ✅ | ✅ | ✅ | 85,88,92,95–99 |
| | ✅ GC level $\gamma_s^{\mathrm{GC}} = 1$ | ✅ GC level $\gamma_s^{\mathrm{GC}} \approx 1$ | ✅ GC level $\gamma_s^{\mathrm{C}} < 1$ | 85,86,89,90,100,101 93,94,102–104 |
| | $R_{\mathrm{out}}/R_{\mathrm{side}} \approx 1$ | $R_{\mathrm{out}}/R_{\mathrm{side}} \lesssim 1$ | $R_{\mathrm{out}}/R_{\mathrm{side}} \lesssim 1$ | 105–113 |
| Direct photons at low $p_{\mathrm{T}}$ | ✅ | ? | ❌ | 114–116 |
| $v_{\mathrm{n}}$ in events with Z, jets | ? | ✅ up to $v_3$ | ✅ $v_2$ | 117–119 |
| Jet constituent $v_{\mathrm{n}}$ | ✅ $v_2$ | ✅ $v_2$ | ✅ $v_2$ in jet frame | 120,121 |
| **Hard probes** | ✅ | ❌ | ❌ | 122–139 |
| | ? | ✅ | ? | ? |
| … through dijet asymmetry | ✅ | ❌ | ❌ | 140–147 |
| … through correlations | ✅ ($Z$–jet, $\gamma$–jet, h–jet) | ❌ (h–jet, jet–h) | ? | 139,148–157 |
| | ✅ | ✅ | ? | 117,158–160 |
| **Heavy-flavor** | ✅ up to $v_3$ (c), up to $v_2$ (b) | ✅ up to $v_2$ | ✅ up to $v_2$ | 40,161–183 |
| | ✅ | ✅ | ? | 167,184–219 |

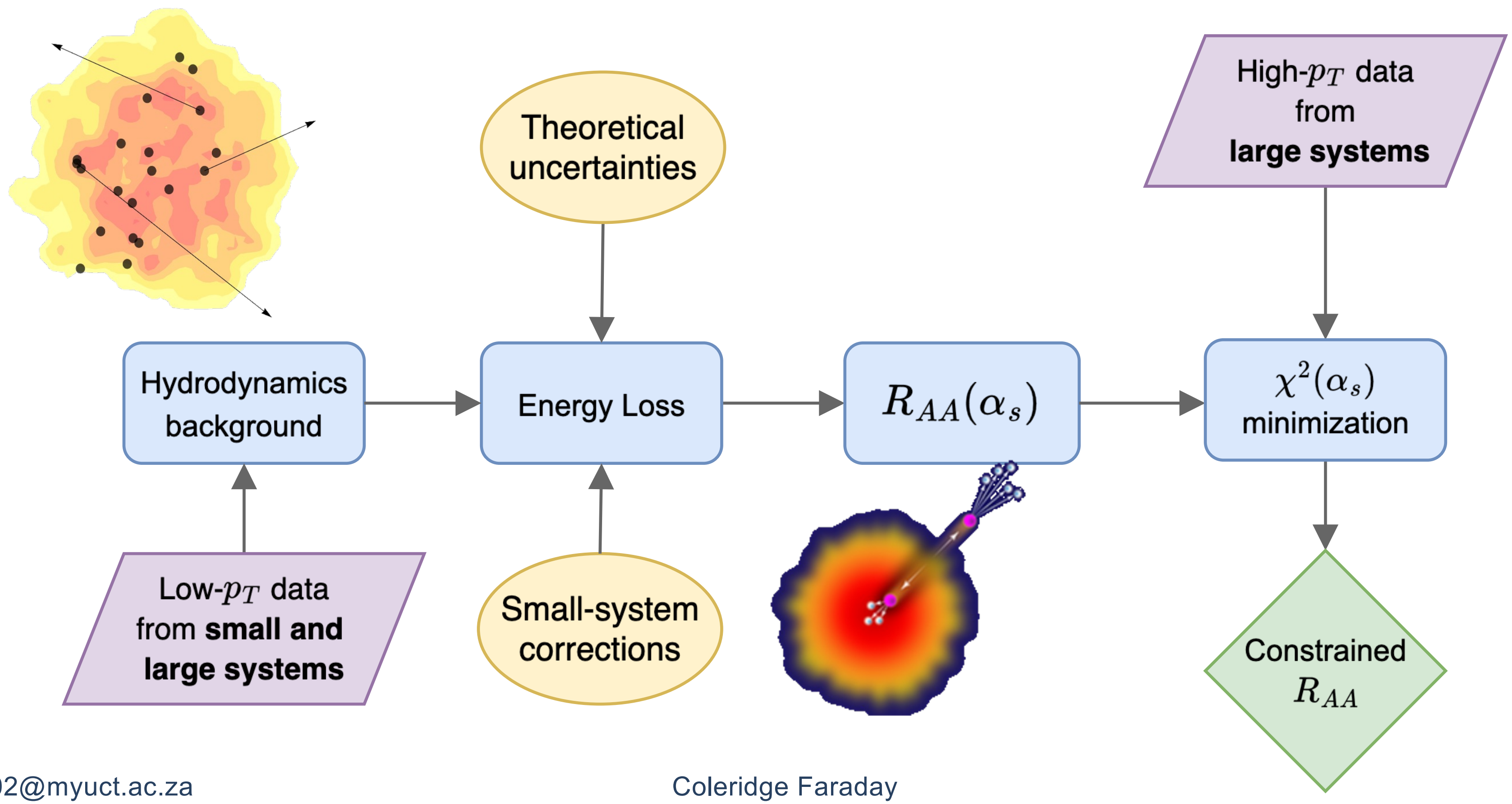
✅ = observed
❌ = not observed
? = not measured

## Mixed messages?

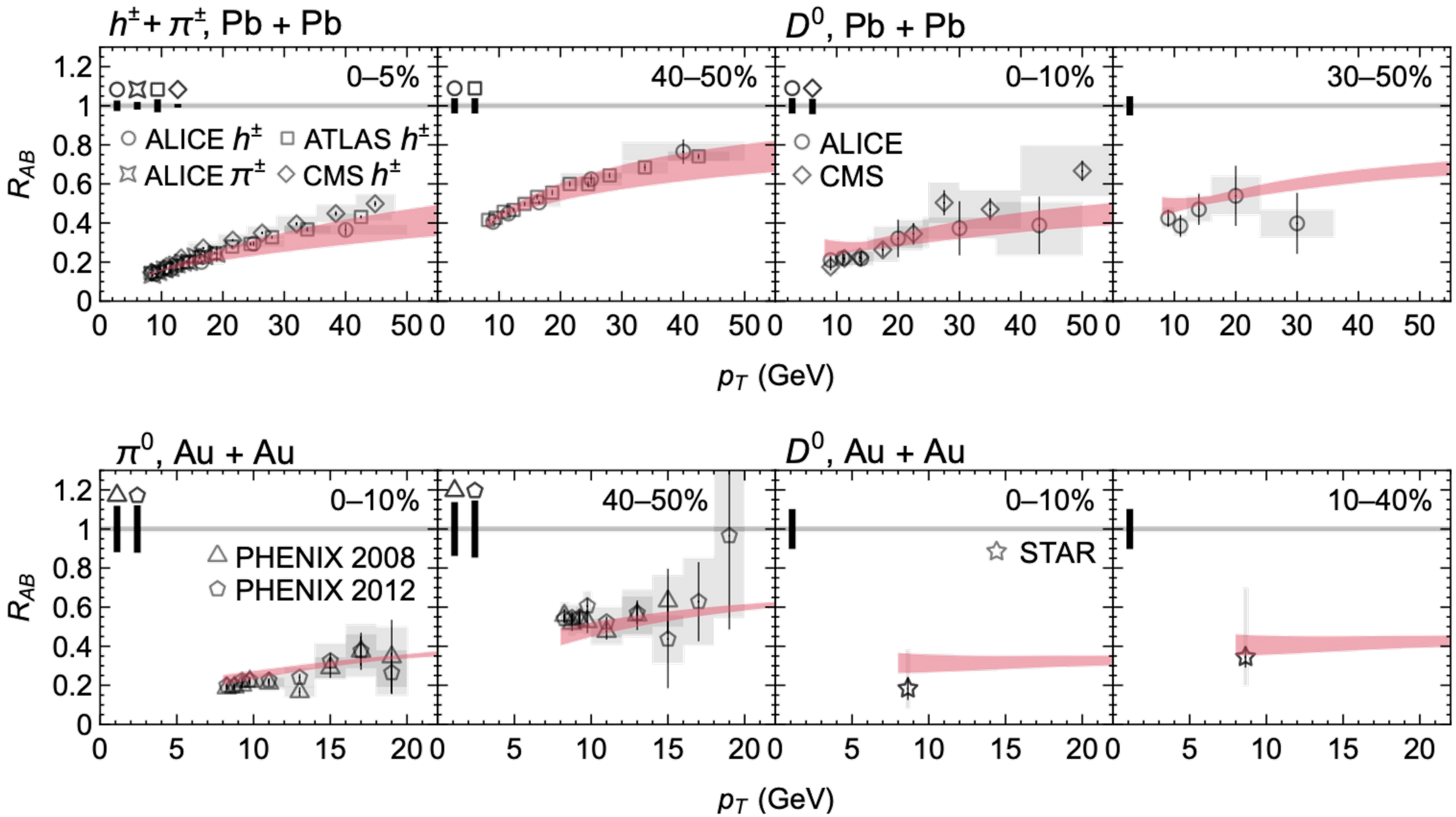
## Hard probes are ***the*** missing piece of the puzzle!

Adapted from J. F. Grosse-Oetringhaus and U. A. Wiedemann, arXiv:2407.07484 [hep-ex] (2024). See backup slides for references

# Our Approach

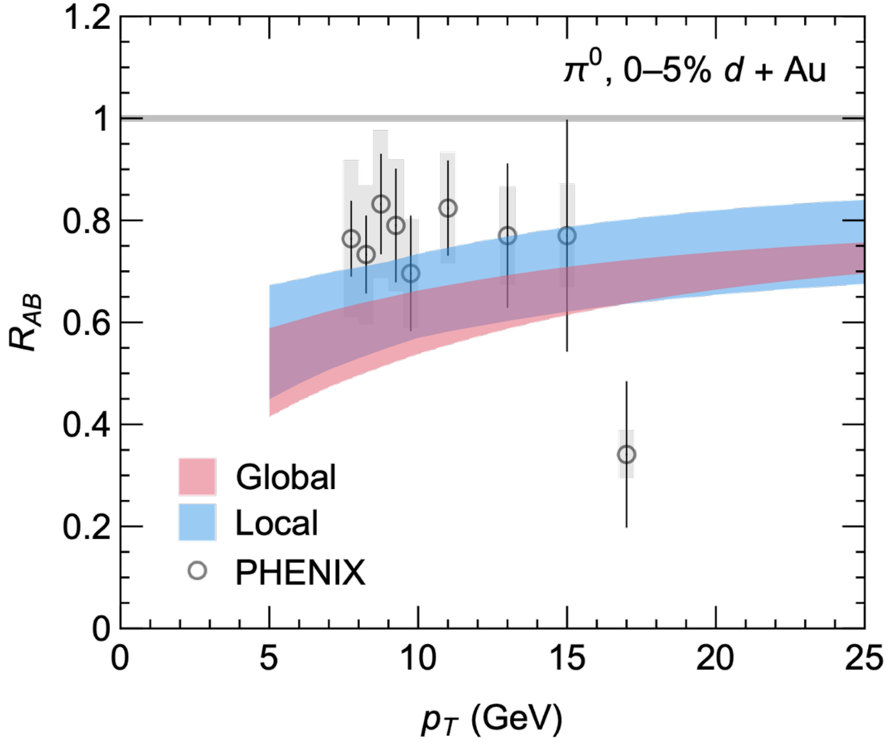


# Results in Large Systems




Good agreement with all available single-inclusive high-$p_T$ data

# Small system suppression at RHIC

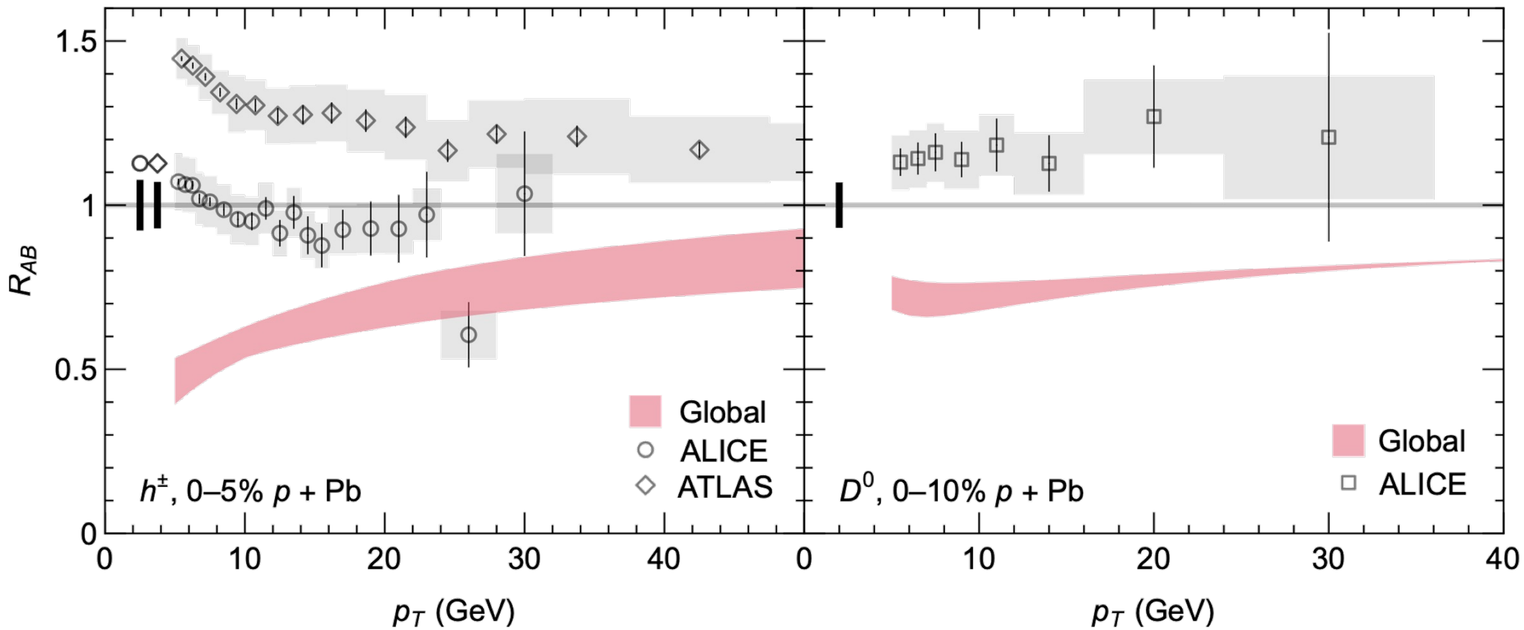



- PHENIX measures $\gamma$-normalized $R_{AB}$ which removes the need to use the Glauber model (aiming to reduce centrality bias)
- Good agreement between our results and data
- Global and local fits agree within uncertainties

*PHENIX, Phys. Rev. Lett.* ***134****, 022302 (2025).*

CF and W. A. Horowitz, arXiv:2505.14568 [hep-ph] (2025).

# Small system suppression at LHC




ALICE, Phys. Rev. C **91**, 064905 (2015).

ALICE, JHEP **12**, 092 (2019).

ATLAS, JHEP **07**, 074 (2023).

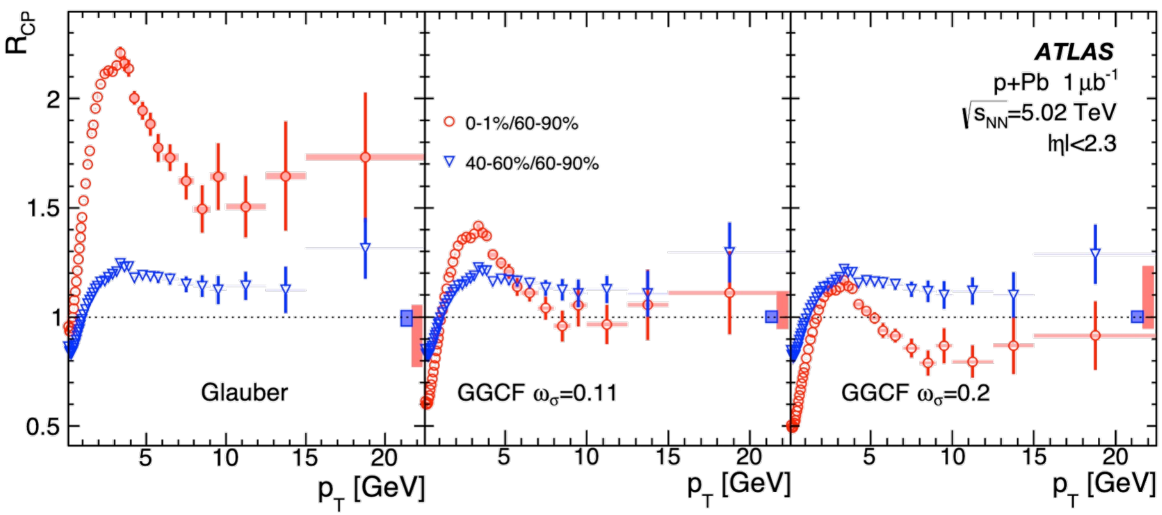
- Strong disagreement with LHC results
- Likely due to selection biases and model dependencies in the measurement of $R_{pA}$!
- Repeat of LHC $\gamma$-normalized measurement would help decrease centrality bias

# Hard / soft correlations

## Is $R_{pA}$ simply a flawed observable?

- Correlations between total number of particles and high momentum particle production have been measured
- Significant Glauber model dependence in determining $N_{coll}$

Open question: do these correlations effect only **centrality categorization** or do they effect the **size and temperature** of the produced QGP?

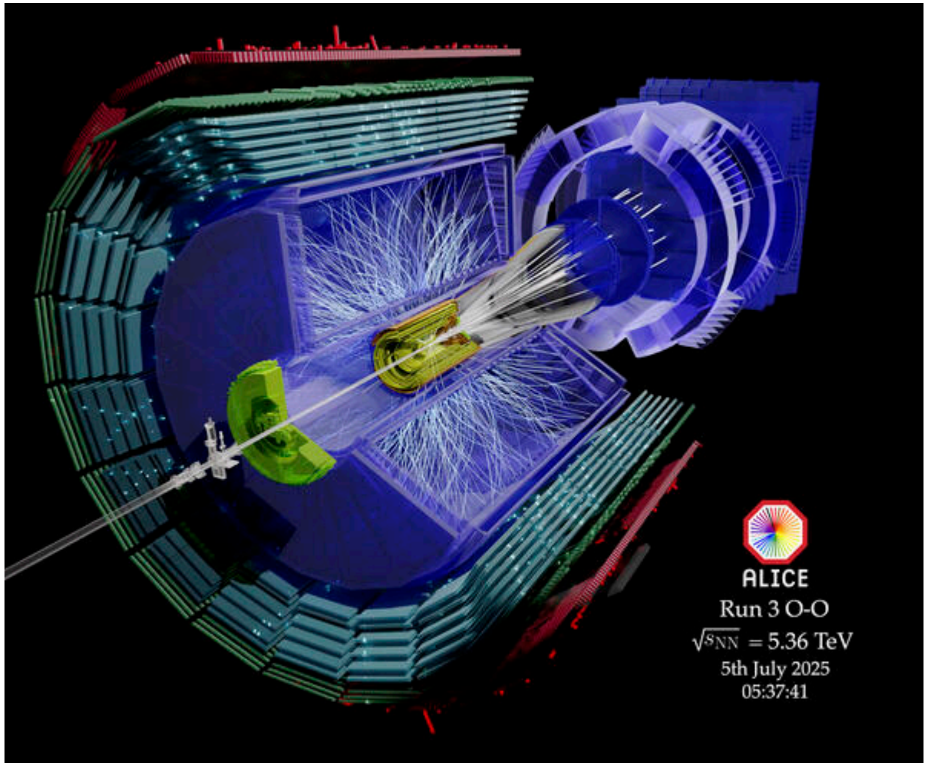


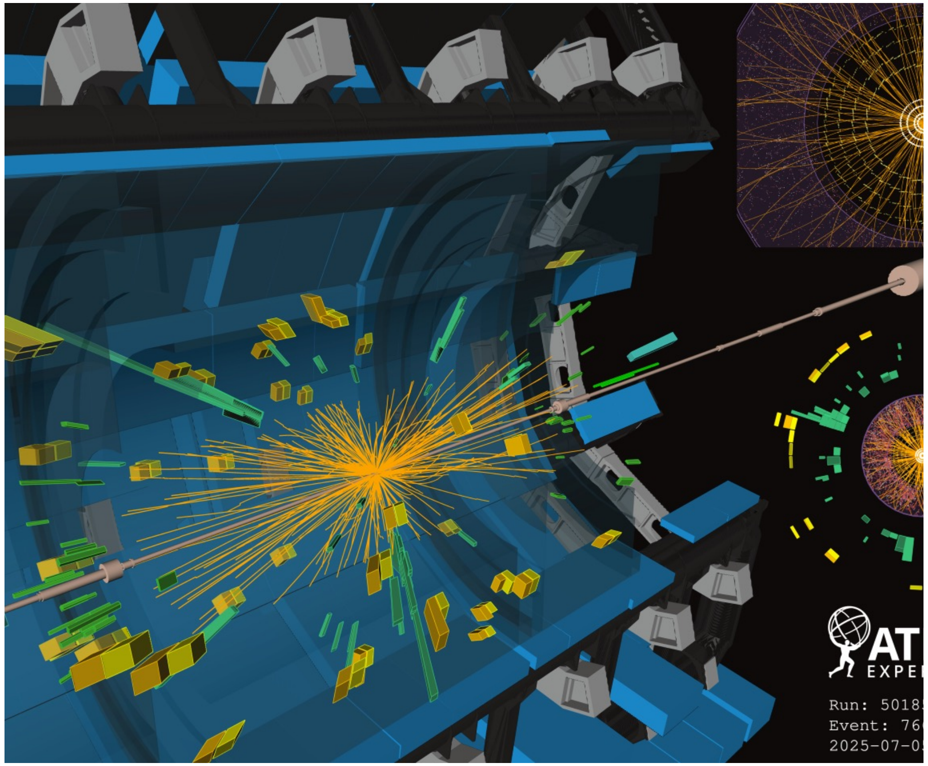

ATLAS, Phys. Lett. B **763**, 313 (2016).

ATLAS, arXiv:2504.02638 [nucl-ex] (2025).

# Oxygen Collisions

First oxygen collisions at LHC finished this week!



# Outlook

- Understanding whether there is energy loss in small systems is **the missing puzzle piece** in understanding QGP formation in small systems
  - Far-reaching implications from the early universe to the onset of many-particle dynamics

- We showed that constraining on large system data leads to good agreement with the model-independent measurement of the $R_{pA}$ in small systems

- Selection biases and model dependences plague centrality-selected events with high-$p_T$ particles (particularly when the Glauber model is used). We need:

  - **Theoretical input**: hard soft correlations, quantitative predictions
  - **Self-normalized observables**
  - **More data:** i.e., just completed oxygen run at LHC!